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DYNAMIC MULTI-DIMENSIONAL MEDIA CONTENT PROJECTION

Abstract

In an embodiment, dynamic multi-dimensional media content is projected. Image data including 2D images depicting view of real-world is acquired. 3D polygons are detected corresponding to objects on a ground of the 2D images. 3D surface model is segmented into 3D segments including ground segments and non-ground segments. The ground segment of the 3D surface model is extracted by removing the one or more non-ground segments from the plurality of 3D segments. Corresponding information between 2D pixel coordinates in the 2D images and 3D surface coordinates of the ground segments is determined by applying a ray casting operation on the 2D images and the ground segment of the 3D surface model. The 3D polygons are projected onto the 3D surface model based on the correspondence information.

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Background/Summary

FIELD

[0001] The embodiments discussed in the present disclosure are related to dynamic multi-dimensional media content projection.

BACKGROUND

[0002] The 3D perspective in object detection may be based on its ability to provide a more complete understanding of real-world scenarios than 2D images. This may especially be useful in situations where a thorough investigation is required, such as a car accident. However, the widespread use of 2D monocular cameras, such as CCTVs, due to their low cost, impedes the full realization of 3D perspective's potential. A significant issue may arise from the previous method of projecting polygons onto the ground. This method may work well when the entire ground is visible to the 2D monocular cameras but fails to project them onto an invisible ground. This especially may be problematic in occlusion situations, where objects such as trees or traffic signals block the view.

[0003] The subject matter claimed in the present disclosure is not limited to embodiments that solve any disadvantages or that operate only in environments such as those described above. Rather, this background is only provided to illustrate one example technology area where some embodiments described in the present disclosure may be practiced.

SUMMARY

[0004] According to an aspect of an embodiment, a method may include a set of operations which may include acquiring image data that may include 2D images depicting a view of a real-world location. The set of operations may further include detecting 3D polygons corresponding to objects on a ground region of the 2D images and acquisition of a 3D surface model of the real-world location. The 3D surface model may be segmented into multiple 3D segments comprising a ground segment and one or more non-ground segments. The ground segment of the 3D surface model may be extracted by removing the non-ground segments from a plurality of 3D segments. The set of operations may further include determining, by applying a ray casting operation on the 2D images and the ground segment of the 3D surface model, correspondence information between 2D pixel coordinates in the 2D images and 3D surface coordinates of the ground segment and projecting, based on the correspondence information, 3D polygons corresponding to the objects onto the 3D surface model.

[0005] The objects and advantages of the embodiments will be realized and achieved at least by the elements, features, and combinations particularly pointed out in the claims.

[0006] Both the foregoing general description and the following detailed description are given as examples and are explanatory and are not restrictive of the invention, as claimed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] Example embodiments will be described and explained with additional specificity and detail through the use of the accompanying drawings in which:

[0008] FIG. **1** is a diagram representing an example environment related to dynamic multi-dimensional media content projection;

[0009] FIG. **2** is a block diagram that illustrates an exemplary electronic device for dynamic multi-dimensional media content projection;

[0010] FIG. **3**A is a diagram that illustrates an execution pipeline for object detection and ground segmentation in dynamic multi-dimensional media content projection;

[0011] FIG. **3**B is a diagram that illustrates an execution pipeline for determining correspondence information for dynamic multi-dimensional media content projection;

[0012] FIG. **4** is a diagram that illustrates a flowchart of an example method for estimation of 3D polygons corresponding to objects in 2D images;

[0013] FIG. **5** is a diagram that illustrates an exemplary scenario for detection and estimation of 3D polygons;

[0014] FIG. **6** is a diagram that illustrates an exemplary scenario for acquisition of 3D surface model;

[0015] FIG. **7** is a diagram that illustrates a flowchart of an example for determination of correspondence information for dynamic multi-dimensional media content projection;

[0016] FIG. **8** is a diagram that illustrates an exemplary scenario for estimation of initial camera pose parameters corresponding to an initial view of the 3D surface model in a 3D space;

[0017] FIG. **9** is a diagram that illustrates a flowchart of an example for dynamic multi-dimensional media content projection,

[0018] all according to at least one embodiment described in the present disclosure.

DESCRIPTION OF EMBODIMENTS

[0019] Some embodiments described in the present disclosure relate to methods and systems for dynamic multi-dimensional media content projection. In the present disclosure, image data may be acquired. The image data may include 2-Dimensional (2D) images depicting a view of a real-world location. Thereafter, 3D polygons (e.g., a bounding box) may be detected corresponding to objects on a ground region of the 2D images. A 3-Dimensional (3D) surface model of the real-world location may be acquired for segmenting into a plurality of 3D segments include ground segments and one or more non-ground segments. Based on the segmented 3D surface model, the ground segment of the 3D surface model may be extracted by removing the one or more non-ground segments from the plurality of 3D segments. Correspondence information between 2D pixel coordinates in the 2D images and 3D surface coordinates of the ground segment may be determined, by applying a ray casting operation on the 2D images and the ground segment. One or more 3D polygons may be projected corresponding to the objects onto the 3D surface model.

[0020] According to one or more embodiments of the present disclosure, the technological field of 3D image generation and projection may be improved by configuring an electronic device in a manner that the electronic device is able to dynamically project multi-dimensional media content using correspondence information between 2D and 3D space coordinates. The electronic device may acquire image/video data that includes 2D images depicting a view of the real-world location. Further, the electronic device may detect 3D polygons corresponding to objects on a ground region of the 2D images. Based on the acquired 3D surface model of the real-world location, the 3D surface model may be segmented into the plurality of 3D segments comprising a ground segment and non-ground segments. The ground segment of the 3D surface model may be extracted by removing the non-ground segments from the plurality of 3D segments. Further, the electronic device may determine correspondence information between the 2D pixel coordinates in the 2D images and the 3D surface coordinates of the ground segment by applying the ray casting operation on the 2D images image and the ground segment. Based on the correspondence information, the electronic device may project the 3D polygons corresponding to the objects onto the 3D surface model.

[0021] Conventional techniques of 3D image projections may provide some benefits by comprehending real-world situations beyond the limitations of 2D images. The 3D perspectives may assist, for example, to visualize a vehicle accident from multiple angles to identify potential issues. It may be appreciated that generation of a 3D model from the 2D images may involve capturing the 2D images from different perspectives of the objects or an environment. The 2D images may be processed using a 3D model generation component, which generates 3D models based on the 2D images and the 3D data associated with them. Alternatively, a laser-based 3D scanning may be used to directly acquire a 3D surface model.
[0022] In conventional 3D object detection, a significant issue may arise in the approach of projecting polygons (e.g., bounding cubes) onto a ground surface. The method may work well when an entire ground is visible from an imaging device (e.g., 2D imaging device) but may fail to project them onto an invisible ground. This becomes particularly problematic in occlusion situations, where objects like trees or traffic signals may obstruct the view. The main challenge is to project such polygons onto a precise ground position, which may be required for accurate object detection. A sub-issue may arise when the polygons are unavailable or incorrectly detected, complicating the task even further. As a result, while the 3D perspectives may have the potential to improve object detection, these issues must be addressed before it can fully realize its benefits. In contrast, the present disclosure provides an approach which can precisely project polygons, from the 2D image to 3D space by introducing a correspondence table between the 2D pixel coordinates in the 2D images and the 3D surface coordinates, even if there are obstacles and the objects are partially obscured or invisible.
[0023] Embodiments of the present disclosure are explained with reference to the accompanying drawings.
[0024] FIG. **1** is a diagram representing an example environment related to dynamic multi-dimensional media content projection, arranged in accordance with at least one embodiment described in the present disclosure. With reference to FIG. **1**, there is shown an environment **100**. The environment **100** may include an electronic device **102**, a server **104**, a database **106**, a communication network **108**, a 2D imaging device **110**, and a 3D imaging device **112**. The electronic device **102**, the server **104**, the 2D imaging device **110**, and the 3D imaging device **112** may be communicatively coupled to each other, via the communication network **108**. The electronic device **102** may include a display device **102**A. The server **104** may be communicatively coupled to a database **106**. In FIG. **1**, there is further shown a set of images **110**A acquired by the 2D imaging device **110** and a 3D surface model **112**A acquired by the 3D imaging device **112**. The set of images (for example 2D images **110**A and the 3D surface model **112**A) may be stored in the database **106**.
[0025] The set of images (for example, 2D images **110**A) and 3D surface model **112**A in FIG. **1** are presented merely as an example. The set of images may include N number of images, without deviation from the scope of the disclosure.
[0026] The 2D imaging device **110** (e.g., camera) and the 3D imaging device **112** (e.g., LIDAR) may be communicatively coupled to the electronic device **102**. In some embodiments, the electronic device **102** may include the 2D imaging device **110** and the 3D imaging device **112**.
[0027] The electronic device **102** may include suitable logic, circuitry, interfaces, and/or code that may be configured to project multi-dimensional media content, as described herein. In certain embodiments, the electronic device **102** may be configured to acquire the 2D images **110**A and 3D surface model **112**A, as shown in FIG. **1**. Examples of the multi-dimensional media content may include, but are not limited to, 2D images **110**A, 3D surface model **112**A, 2D video, and 3D video. Examples of the electronic device **102** may include, but are not limited to, a mobile device, a desktop computer, a laptop, a computer workstation, an action camera, 360 cameras, a computing device, a mainframe machine, a gaming console, a virtual reality (VR) device, an augmented reality (AR) device, a mixed reality (MR) device, a mainframe machine, a computer workstation, an

internet of things (IoT) device. In one or more embodiments, the electronic device **102** may include a user-end terminal device and a server communicatively coupled to the user-end terminal device. The electronic device **102** may be implemented using hardware including a processor or a microprocessor (e.g., to perform or control performance of one or more operations). In some other instances, the electronic device **102** may be implemented using a combination of hardware and software.

[0028] The server **104** may include suitable logic, interfaces, and/or code that may be configured to store neural network models (such as the object detection model, polygon estimation model **208**B, 3D surface model **112**A). The server **104** may be configured to retrieve from the database **106**, historical data associated with the 3D polygons corresponding to objects in the 2D images **110**A or the 3D surface model **112**A. Each 3D polygon may enclose a corresponding object of the plurality of objects. The server **104** may be configured to estimate 3D polygons corresponding to the detected one or more objects in the 2D images **110**A, based on the trained polygon estimation model **208**B.

[0029] The database **106** may be stored or cached on a device, such as a server or the electronic device **102**. The device storing the database **106** may be configured to receive a query for the data (e.g., the 2D images **110**A and the 3D surface model **112**A) from the electronic device **102**. In response, the electronic device **102** of the database **106** may be configured to retrieve and provide the queried data to the electronic device **102** based on the received query. In some embodiments, the database **106** may be hosted on a plurality of servers stored at the same or different locations. The operations of the database **106** may be executed using hardware including a processor, a microprocessor (e.g., to perform or control performance of one or more operations), a field-programmable gate array (FPGA), or an application-specific integrated circuit (ASIC). In some other instances, the database **106** may be implemented using software.

[0030] The communication network **108** may include a communication medium through which the electronic device **102** may communicate with the server(s) or device(s) that may store the database **106**, and the imaging devices (e.g., 2D imaging device **110** and 3D imaging device **112**). Examples of the communication network **108** may include, but are not limited to, the Internet, a cloud network, a Wireless Fidelity (Wi-Fi) network, a Personal Area Network (PAN), a Local Area Network (LAN), a cellular network (such as, a Long-term evolution (or 4G) cellular network or a 5G cellular network), a satellite network (such as a network of low earth orbit satellites), and/or a Metropolitan Area Network (MAN)). Various devices in the environment **100** may be configured to connect to the communication network **108**, in accordance with various wired and wireless communication protocols. Examples of such wired and wireless communication protocols may include, but are not limited to, at least one of a Transmission Control Protocol and Internet Protocol (TCP/IP), User Datagram Protocol (UDP), Hypertext Transfer Protocol (HTTP), File Transfer Protocol (FTP), ZigBee, EDGE, IEEE 802.11, light fidelity (Li-Fi), 802.16, IEEE 802.11s, IEEE 802.11g, multi-hop communication, wireless access point (AP), device to device communication, cellular communication protocols, and/or Bluetooth (BT) communication protocols, or a combination thereof.

[0031] The server **104**, the 2D imaging device **110**, and the 3D imaging device **112** may be communicatively coupled via a wired or wireless network. In some embodiment, the electronic device **102** may incorporate the functionality of the server **104**, the 2D imaging device **110**, and the 3D imaging device **112**.

[0032] The 2D imaging device **110** may include suitable logic, circuitry, and interfaces that may be configured to capture an image or a plurality of images of a real-world location that includes various objects (e.g., buildings, trees, roads, ground surface, traffic signals, people etc.). The 2D imaging device **110** may be further configured to capture 2D images **110**A corresponding to real world locations. The 2D images **110**A may be any digital data, which can be rendered, streamed, broadcasted, or stored on any electronic device **102** or storage. Examples of the 2D imaging device

110 may include, but are not limited to, an image sensor, a wide-angle camera, an action camera, a closed-circuit television (CCTV) camera, a camcorder, a camera with an integrated depth sensor, a cinematic camera, Digital Single-Lens Reflex (DSLR) camera, a Digital Single-Lens Mirrorless (DSLM) camera, a digital camera, camera phones, a time-of-flight camera (ToF camera), a night-vision camera, and/or other image capture devices.

[0033] The 3D imaging device **112** may include suitable logic, circuitry, and interfaces that may be configured to capture 3D data by performing a 3D scan of real-world objects (e.g., buildings, trees, roads, ground surface, traffic signals, people etc.). The 3D data (e.g., depth map or point cloud) may be converted to the 3D surface model **112**A. Examples of the 3D imaging device **112** may include, but are not limited to, LiDAR, a time-of-flight camera (ToF camera), stereoscopic camera, structured-light 3D scanner, CT scanner. The 3D surface model **112**A of the real-world location may be acquired from the 3D imaging device **112**.

[0034] The 3D data may be collected from the 3D imaging device **112** to acquire the 3D surface model **112**A of the real-world location (e.g., a street in a city). The 3D data may be any digital data, which can be rendered, streamed, broadcasted, or stored on any electronic device **102** or storage. Examples of the 3D data may include, but are not limited to, depth maps, 3D volumetric data, and point cloud data.

[0035] During operation, the electronic device **102** may acquire image data that includes the 2D images **110**A depicting a view of a real-world location. The electronic device **102** may be configured to detect 3D polygons corresponding to objects on a ground region of the 2D images **110**A. The electronic device **102** may be further configured to acquire the 3D surface model **112**A of a real-world location from 3D data received at the 3D imaging device **112**.

[0036] In at least one embodiment, the electronic device **102** may estimate initial camera pose parameters corresponding to an initial view of the 3D surface model **112**A in a 3D space. The electronic device **102** may optimize the initial camera pose parameters to align a view of the 3D surface model **112**A to a view of the 2D images **110**A.

[0037] The electronic device **102** may be further configured to segment the 3D surface model **112**A into a plurality of 3D segments comprising a ground segment and non-ground segments. The ground segment of the 3D surface model **112**A may be extracted by removing the one or more non-ground segments from the plurality of 3D segments. By applying the ray casting operation on the 2D images **110**A and the ground segment of the 3D surface model **112**A, the electronic device **102** may be configured to determine correspondence information between 2D pixel coordinates in the 2D images **110**A and 3D surface coordinates of the ground segment. By way of example, and not limitation, the correspondence information may be a correspondence table that may include 2D pixel coordinates of the 2D images **110**A and the 3D surface coordinates corresponding to the 2D pixel coordinates. The electronic device **102** may be further configured to project, based on the correspondence information, one or more 3D polygons (e.g., a 3D cube/cuboid) corresponding to the objects onto the 3D surface model **112**A.

[0038] The electronic device **102** may generate vertex information that includes a mapping of the 3D surface coordinates with vertices of the one or more 3D polygons. The electronic device **102** may further generate edge information that includes a mapping of the vertices to edges of the one or more 3D polygons. The projection may be performed based on the vertex information and the edge information. The vertex information may include a primary table that includes a vertex ID, and a polygon ID, a vertex type, and the 3D surface coordinates. The edge information may include a secondary table that includes the polygon ID, an edge ID, and IDs of edge vertices. The vertex type may include one of a ground or an object. Each 3D polygon of the one or more 3D polygons may enclose a corresponding object of the plurality of objects in a 3D space of the 3D surface model **112**A.

[0039] In at least one embodiment, the electronic device **102** may retrieve, from a database, historical data associated with the 3D polygons corresponding to a plurality of objects in a set of images (e.g., 2D images **110**A). Each 3D polygon of the plurality of 3D polygons encloses a corresponding object of the plurality of objects. The electronic device **102** may train the polygon estimation model **208**B based on the historical data. During inference, the electronic device **102** may estimate the 3D polygons corresponding to the detected objects in the 2D images **110**A, based on the trained polygon estimation model **208**B. The electronic device **102** may project the detected 3D polygons onto the ground region of the 2D images **110**A based on a determination that a difference between the estimated 3D polygons and the detected 3D polygons is less than a predefined threshold. Alternatively, the electronic device **102** may project the estimated 3D polygons onto the ground region of the 2D images **110**A based on the determination that a difference between the estimated 3D polygons and the detected 3D polygons is greater than the predefined threshold. Details related to the historical data and the behavioral data are further provided, for example, in FIG. **4** (at **402**, **412**).

[0040] Modifications, additions, or omissions may be made to FIG. **1** without departing from the scope of the present disclosure. For example, the environment **100** may include more or fewer elements than those illustrated and described in the present disclosure. For instance, in some embodiments, the environment **100** may include the electronic device **102** but not the database **106**. In addition, in some embodiments, the functionality of each of the database **106** may be incorporated into the electronic device **102**, without a deviation from the scope of the disclosure.

[0041] FIG. **2** is a block diagram that illustrates an exemplary electronic device **102** for dynamic multi-dimensional media content projection, arranged in accordance with at least one embodiment described in the present disclosure. FIG. **2** is explained in conjunction with elements from FIG. **1**. With reference to FIG. **2**, there is shown a block diagram **200** of the electronic device **102**. The electronic device **102** may include network interface **202**, an input/output (I/O) device **204**, a circuitry **206**, and a memory **208**. The I/O device may include a display device **102**A. The memory may include an object detector **208**A, a polygon estimation model **208**B, a 3D surface model **208**C.

[0042] The network interface **202** may comprise suitable logic, circuitry, interfaces, and/or code that may be configured to establish a communication between the electronic device **102**, the server of the database **106**, the 2D imaging device **110**, and the 3D imaging device **112**, via the communication network **108**. The network interface **202** may be implemented by use of various known technologies to support wired or wireless communication of the electronic device **102**, via the communication network **108**. The network interface **202** may include, but is not limited to, an antenna, a radio frequency (RF) transceiver, one or more amplifiers, a tuner, one or more oscillators, a digital signal processor, a coder-decoder (CODEC) chipset, a subscriber identity module (SIM) card, and/or a local buffer.

[0043] The I/O device **204** may include suitable logic, circuitry, interfaces, and/or code that may be configured to receive a user input. For example, the I/O device **204** may display image data (e.g., the 2D images **110**A) and the 3D surface model **112**A captured using the 2D imaging device **110** and the 3D imaging device **112**, respectively. The I/O device **204** may project the 3D polygons corresponding to the objects onto the 3D surface model **112**A. The I/O device **204** may include various input and output devices, which may be configured to communicate with the circuitry **206** and other components, such as the network interface **202**. Examples of the input devices may include, but are not limited to, a touch screen, a keyboard, a mouse, a joystick, and/or a microphone. Examples of the output devices may include, but are not limited to, a display (e.g., the display device **102**A) and a speaker.

[0044] The circuitry **206** may include suitable logic, circuitry, and/or interfaces that may be configured to execute program instructions associated with different operations to be executed by the electronic device **102**. For example, some of the operations may include the acquisition of the image data, detection of the 3D polygons, segmentation of 3D surface model **208**C, extraction of the ground segment and non-ground segment, determination of correspondence information and projection of the 3D polygons. The circuitry **206** may include any suitable special-purpose or general-purpose computer, computing entity, or processing device including various computer hardware or software modules and may be configured to execute instructions stored on any applicable computer-readable storage media. For example, the circuitry **206** may include a microprocessor, a microcontroller, a digital signal processor (DSP), an application-specific integrated circuit (ASIC), a Field-Programmable Gate Array (FPGA), or any other digital or analog circuitry configured to interpret and/or to execute program instructions and/or to process data.

[0045] Although illustrated as a single processor in FIG. **2**, the circuitry **206** may include any number of processors configured to, individually or collectively, perform or direct performance of any number of operations of the electronic device **102**, as described in the present disclosure. Additionally, one or more of the processors may be present on one or more different electronic devices, such as different servers. In some embodiments, the circuitry **206** may be configured to interpret and/or execute program instructions and/or process data stored in the memory **208**. In some embodiments, the circuitry **206** may fetch program instructions from the memory **208** and load the program instructions in the memory **208**. After the program instructions are loaded into the memory **208**, the circuitry **206** may execute the program instructions. Some of the examples of the circuitry **206** may be a Graphics Processing Unit (GPU), a Central Processing Unit (CPU), a Reduced Instruction Set Computer (RISC) processor, an ASIC processor, a Complex Instruction Set Computer (CISC) processor, a co-processor, and/or a combination thereof.

[0046] The memory **208** may include suitable logic, circuitry, interfaces, and/or code that may be configured to store program instructions executable by the circuitry **206**. In certain embodiments, the memory **208** may be configured to store operating systems and associated application-specific information. The memory **208** may include computer-readable storage media for carrying or having computer-executable instructions or data structures stored thereon. Such computer-readable storage media may include any available media that may be accessed by a general-purpose or special-purpose computer, such as the circuitry **206** By way of example, and not limitation, such computer-readable storage media may include tangible or non-transitory computer-readable storage media including Random Access Memory (RAM), Read-Only Memory (ROM), Electrically Erasable Programmable Read-Only Memory (EEPROM), Compact Disc Read-Only Memory (CD-ROM) or other optical disk storage, magnetic disk storage or other magnetic storage devices, flash memory devices (e.g., solid state memory devices), or any other storage medium which may be used to carry or store particular program code in the form of computer-executable instructions or data structures and which may be accessed by a general-purpose or special-purpose computer. Combinations of the above may also be included within the scope of computer-readable storage media. Computer-executable instructions may include, for example, instructions and data configured to cause the circuitry **206** to perform a certain operation or group of operations associated with the electronic device **102**. The memory **208** may include various models, but not limited to the object detector **208**A, the polygon estimation model **208**B, the 3D surface model **208**C. The various models are described in detail in FIG. **3**, FIG. **5**, FIG. **8**, and FIG. **9**.

[0047] By way of example, and not limitation, such computer-readable storage media may include tangible or non-transitory computer-readable storage media including Compact Disc Read-Only Memory (CD-ROM) or other optical disk storage, magnetic disk storage or other magnetic storage devices (e.g., Hard-Disk Drive (HDD)), flash memory devices (e.g., Solid State Drive (SSD), Secure Digital (SD) card, other solid state memory devices), or any other storage medium which may be used to carry or store particular program code in the form of computer-executable instructions or data structures and which may be accessed by a general-purpose or special-purpose computer. Combinations of the above may also be included within the scope of computer-readable storage media. Computer-executable instructions may include, for example, instructions and data configured to cause the circuitry **206** to perform a certain operation or group of operations associated with the electronic device **102**.

[0048] The object detection (using the object detector **208**A) may be achieved through semantic segmentation, where different regions in the image (e.g., 2D image **110**A) are defined and labeled based on the real-life objects (e.g., vehicle, people, buildings etc.) they depict. The resulting objects detection may then be used to generate point features that may be placed on the map. In an exemplary scenario, a user walking on a footpath may appear in the image (e.g., 2D image **110**A), The user may be considered as the object. The objects may be static objects or dynamic objects. In a dynamic sequence of images, few of the images may include occlusions which partially or fully occlude the object. The object detection may be useful for many applications, such as but not limited to, security, surveillance, face recognition, autonomous driving, and robotics. The circuitry **206** may detect and track pedestrians, vehicles, and traffic signs on a busy street. This can help improve road safety, traffic management, and navigation. Based on the object detection in 2D images **110**A, algorithm may be enabled for differentiating objects from background. The polygons may also handle occlusion, as they can accurately label only the visible parts of occluded objects.

[0049] The polygon estimation model **208**B may be a neural network or computational network or a system of artificial neurons, arranged in a plurality of layers, as nodes. During inference, the polygon estimation model **208**B may process input images to localize objects across the 2D images (e.g., the 2D images **110**A) by extracting behavioral data associated with movement or appearance of such objects. The plurality of layers of the neural network may include an input layer, one or more hidden layers, and an output layer. Each layer of the plurality of layers may include one or more nodes (or artificial neurons, represented by circles, for example). Outputs of all nodes in the input layer may be coupled to at least one node of hidden layer(s). Similarly, inputs of each hidden layer may be coupled to outputs of at least one node in other layers of the neural network. Outputs of each hidden layer may be coupled to inputs of at least one node in other layers of the neural network. Node(s) in the final layer may receive inputs from at least one hidden layer to output a result. The number of layers and the number of nodes in each layer may be determined from hyper-parameters of the neural network. Such hyper-parameters may be set before or after training the neural network on a training dataset.

[0050] Each node of the neural network in the polygon estimation model **208**B may correspond to a mathematical function (e.g., a sigmoid function or a rectified linear unit) with a set of parameters, tunable during training of the network. The set of parameters may include, for example, a weight parameter, a regularization parameter, and the like. Each node may use the mathematical function to compute an output based on one or more inputs from nodes in other layer(s) (e.g., previous layer(s)) of the neural network. All or some of the nodes of the neural network may correspond to same or a different same mathematical function.

[0051] In training of the neural network of the polygon estimation model **208**B, one or more parameters of each node of the neural network may be updated based on whether an output of the final layer for a given input (from the training dataset) matches a correct result based on a loss function for the neural network. The above process may be repeated for same or a different input until a minima of loss function is achieved, and a training error is minimized. Several methods for training are known in art, for example, gradient descent, stochastic gradient descent, batch gradient descent, gradient boost, meta-heuristics, and the like.

[0052] Examples of the polygon estimation model **208**B may be a neural network based model, such as but are not limited to, a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a CNN-recurrent neural network (CNN-RNN), R-CNN, Fast R-CNN, Faster R-CNN, an artificial neural network (ANN), (You Only Look Once) YOLO network, a Long Short Term Memory (LSTM) network based RNN, CNN+ANN, LSTM+ANN, a gated recurrent unit (GRU)-based RNN, a fully connected neural network, a Connectionist Temporal Classification (CTC) based RNN, a deep Bayesian neural network, a Generative Adversarial Network (GAN), and/or a combination of such networks. In certain embodiments, the neural network-based polygon estimation model **208**B may be based on a hybrid architecture of multiple Deep Neural Networks (DNNs).

[0053] The 3D surface model **208**C may be a digital representation of features, either real or hypothetical, in three-dimensional space. Some examples of 3D surfaces may be landscape, an urban corridor, gas deposits under the earth, and a network of well depths and the like to determine water table depth. The 3D surface model **208**C can be created from a variety of data sources, such as points, lines, polygons, or images, using interpolation or triangulation methods. The 3D surface model **208**C can be used for various applications, such as determining mass properties, checking interference, generating cross-sections, and creating finite element meshes and the like.
[0054] In some embodiments, the 3D surface model **208**C may be obtained from the 3D imaging device **112** (e.g., LiDAR) or the 3D surface may be generated based on the point cloud data (for example, **602**). The captured point cloud data may be converted to a 3D surface model **208**C (or 3D mesh with or without color/texture).
[0055] The display device **102**A may comprise suitable logic, circuitry, interfaces, and/or code that may be configured to display the information of the image data (e.g., the 2D images **110**A) and the 3D data captured using the 2D imaging device **110** and the 3D imaging device **112**, respectively. The display screen **212** may be configured to receive the user inputs (e.g., for projection of the 3D surface model **112**A) from the user **114**. In such cases the display device **102**A may be a touch screen to receive the user inputs. The display device **102**A may be realized through several known technologies such as, but not limited to, a Liquid Crystal Display (LCD) display, a Light Emitting Diode (LED) display, a plasma display, and/or an Organic LED (OLED) display technology, and/or other display technologies.
[0056] Modifications, additions, or omissions may be made to the example electronic device **102** without departing from the scope of the present disclosure. For example, in some embodiments, the example electronic device **102** may include any number of other components that may not be explicitly illustrated or described for the sake of brevity. The object detector **208**A, polygon estimation model **208**B, and 3D surface model **208**C are described in FIG. **3**, FIG. **4**, FIG. **5**, and FIG. **9**.
[0057] FIGS. **3**A and **3**B are diagrams that collectively illustrate an execution pipeline for dynamic multi-dimensional media content projection, in accordance with an embodiment of the disclosure. FIG. **3**A and FIG. **3**B are described in conjunction with elements from FIG. **1** and FIG. **2**. With reference to FIG. **3**A and FIG. **3**B, there is shown an execution pipeline **300**. The exemplary execution pipeline **300** may include a set of operations that may be executed by one or more electronic components, such as the electronic device **102** of FIG. **1**. The operations may include image data (e.g., 2D images **110**A) acquisition, 3D model acquisition (e.g., 3D surface model **112**A), polygons detection (e.g., 3D polygons), 3D surface model segmentation, ground segment extraction, correspondence information determination, and 3D polygon projection. The set of operations may be performed by the electronic device **102** for dynamic multi-dimensional media content projection, as described herein.
[0058] At **302**, an operation for 2D image acquisition may be executed. In an embodiment, the circuitry **206** may be configured to receive image data that includes the 2D images **110**A. The 2D images **110**A may be any digital data, which can be rendered in 2D, streamed, broadcasted, or stored on any electronic device **102**. For example, the circuitry **206** may receive the image data from the imaging device (for example, the 2D imaging device **110**). The 2D images **110**A may include objects, such as vehicles, ground surface, buildings, roads, poles, and people. In an embodiment, the 2D images **110**A may be pre-stored in the memory **208** and the circuitry **206** may retrieve the pre-stored image data **110**A from the memory **208**.
[0059] At **304**, an operation for 3D polygon detection may be executed. The objects may be static objects, dynamic objects, or a combination thereof. As part of the operation, the circuitry **206** may detect one or more 3D polygons (e.g., a 3D polygon **304**B) corresponding to one or more objects (e.g., a car) on a ground region of the 2D images **110**A. By way of example, and not limitation, the 3D polygon **304**B may be a bounding cube/cuboid which may enclose an object in the images

(such as a 2D images shown in **304**A). Alternatively, the 3D polygons **304**B may be of any suitable shape, without a departure from the scope of the present disclosure.

[0060] In an embodiment, the detection may be performed using a neural network-based object detector (e.g., the object detector **208**A) that may identify and label different regions in the 2D images **304**A based on real-life objects included in such regions. In order to localize the objects, a 3D polygon **304**B may be placed around each object. As an example, the 2D images **110**A of a street may be considered. The detected objects may be, for example, streetlights, vehicles, people, buildings, traffic signals, roads, trees, and the like. The resulting object detections may be used to generate features that may be placed on the map. Consider an exemplary scenario of the 2D images **304**A where a vehicle appears parked in a street. The vehicle may be considered as the object that may be detected by the object detector **208**A. The 3D polygon **304**B (a 3D cube, for example) may be placed to enclose the vehicle based on results generated by the object detector **208**A. In some instances, the 2D images **304**A may also include occlusions (not shown) which may fully or partially occlude the detected object.

[0061] At **306**, an operation for the 3D surface model **112**A acquisition may be executed. In an embodiment, the circuitry **206** may be configured to receive the 3D surface model **112**A that includes 3D mesh or point data. The 3D surface model **112**A may be any digital content, which can be rendered in 3D, streamed, broadcasted, and stored on any electronic device. The 3D surface model **112**A may be generated from 3D data, such as depth maps, 3D volumetric data, and point cloud data.

[0062] The circuitry **206** may receive the 3D data from the imaging device (for example 3D imaging device **112**), which may correspond to objects in the 2D images **110**A. The objects may be vehicles, ground surface, buildings, people, and the like. In another embodiment, the 3D data may be collected from the 3D imaging device **112** and processed to acquire the 3D surface model **112**A of the real-world location (e.g., a street in a city). In another embodiment, the 3D data may be pre-stored in the memory **208** and the circuitry **206** may retrieve the pre-stored 3D data from the memory **208**.

[0063] At **308**, an operation for view alignment may be executed. The view alignment of the acquired 3D surface model **112**A may include estimation of initial pose parameters corresponding to an initial view of the 3D surface model **112**A in the 3D space. The initial camera pose parameters may be optimized to align the view of the 3D surface model **112**A to a view of the 2D images **110**A. The real-world locations and the 3D surface model **112**A may have different coordinates. The real-world locations use longitude and latitude, while the 3D surface model **112**A uses x, y, z axis. This makes it hard to get the precise location and orientation of a fixed camera. Therefore, in order to align the views of the 2D images **110**A and 3D surface model **112**A, values such as location, angle, and zoom may be adjusted. The initial pose parameters corresponding to initial view of the 3D surface model **112**A may be optimized to align the view of the 3D surface model **112**A in the 3D space to obtain aligned 3D surface model **308**A. The aligned 3D surface model **308**A may include the ground segment **308**B and non-ground segments **308**C. For example, the aligned 3D surface model **308**A shows a set of buildings (i.e., the non-ground segments **308**C) as part of the 3D surface model **112**A. The camera pose estimation is further described in FIG. **8**.

[0064] At **310**, the aligned 3D surface model **308**A may be segmented into 3D segments including ground segment **308**B and non-ground segments **308**C. The segmentation may be performed on the aligned 3D surface model **308**A by dividing the aligned 3D surface model **308**A into regions or segments based on criteria, such as color, texture, shape, or semantic meaning (semantic segmentation). In an embodiment, the circuitry **206** may identify and label the objects or parts of a 2D image. For example, the object detector **208**A may locate and identify the objects in the image by drawing bounding boxes around them and assigning class labels, such as ‘car’, ‘person’, ‘road’, or ‘dog’. Based on the labeled objects or parts of the image, the aligned 3D surface model **308**A may be segmented as the ground segment **308**B and non-ground segments **308**C.

[0065] The semantic segmentation of the aligned 3D surface model **308**A may include assignment of a class label to each point in the aligned 3D surface model **308**A, such as ‘car’, ‘road’, ‘sky’, etc. The semantic segmentation may not differentiate between instances of the same class, such as multiple cars in the same image. The instance segmentation may distinguish between instances of the same class, such as multiple cars in the same aligned 3D surface model **308**A. The panoptic segmentation may combine semantic and instance segmentation by assigning a class label and an instance ID to each point on the aligned 3D surface model **308**A, but also grouping points that belong to the same semantic region, such as ‘road 1’ or ‘sidewalk 1’. This type of segmentation may provide a comprehensive and coherent representation of the aligned 3D surface model **308**A.
[0066] In some embodiments, various techniques, and algorithms, such as but not limited to, thresholding, clustering, edge detection, region growing, graph-based methods, or deep learning models may be used to perform segmentation. The deep learning models, such as but not limited to, convolutional neural networks (CNNs), U-Net, Mask R-CNN, DeepLab, and so on. The CNN architecture that consists of an encoder-decoder structure, where the encoder may reduce the spatial resolution of the input image and extracts high-level features, while the decoder may increase the spatial resolution and may generate the output segmentation map.
[0067] At **312**, extraction of a ground segment **308**B from the aligned 3D surface model **308**A may be performed by removing the non-ground segments **308**C from the 3D segments. For example, a segmented image **312**A depicts a ground segment **312**B (i.e., roads).
[0068] At **314**, the ray casting operation may be performed. The ray casting operation may be performed on the 2D images **110**A and the ground segment **308**B of the 3D surface model **112**A. Using the 3D surface model **112**A, each pixel of a 2D image may be assigned a 3D coordinate in 3D space based on the perspective projection formula. As an example, the image size of the 2D images **110**A may be extracted and a 2D coordinate may be selected from 2D pixel coordinates. The ray casting operation may be applied on the selected 2D coordinate to estimate a corresponding 3D surface coordinate on the ground segment **308**B in the 3D space. The above operation may be repeated for all the pixels of the 2D image. The correspondence information may be determined based on the 2D pixel coordinate and the 3D surface coordinate.
[0069] At **316**, correspondence information between the 2D pixel coordinates in the 2D images **110**A and 3D surface coordinates of the ground segment **308**B may be determined based on application of the ray casting operation on the 2D images **110**A and the ground segment **308**B of the 3D surface model **112**A. In an embodiment, the correspondence information may be stored in the database **106**.
[0070] In an example embodiment, the correspondence information may be a table **316**A that includes the 2D pixel coordinates and the 3D surface coordinates corresponding to the 2D pixel coordinates. The correspondence table **316**A may be used for projection of 3D polygons (e.g., a 3D cube around a 3D object) on the surface of the 3D surface model **112**A. In the correspondence table **316**A, the 2D pixel coordinates may include, for example, the width (W.sub.2D) and height (h.sub.2D) parameters (as shown). The 3D pixel coordinates may include the 3D information, for example, the horizontal distance (X.sub.3D), vertical distance (Y.sub.2D), and depth distance (Z.sub.2D) parameters. The camera coordinate system or 2D pixel coordinates may be represented as b.sub.i∈custom-character.sup.2 and the 3D pixel coordinates may be represented as a.sub.i∈custom-character.sup.3.
[0071] At **318**, an operation of polygon shaping may be executed. The polygons **318**C (for example, 2D polygons or 3D polygons **318**C) may include vertices and edges. The polygon shaping representation may include a type of projection with ground and non-ground vertices. For instance, the polygons **318**C may be generated based on the ground vertices. Table. 1 represents a vertices table. The vertices table may include, for example, polygon ID, vertex ID, vertex type, and 3D pixel coordinates. Table. 2 represents an edge table. The edges may include, for example, polygon ID, Edge ID, vertex FROM ID, vertex TO ID. The vertex table may be referred as a primary table

and the edge table may be referred as a secondary table.
TABLE-US-00001 TABLE 1 Polygon ID Vertex ID Vertex Type x.sub.3D y.sub.3D z.sub.3D 0 0 ground 0.1315 0.0235 0.5105 0 1 ground 0.1315 0.0235 0.5131 0 2 object 0.6336 0.0315 0.3351
TABLE-US-00002 TABLE 2 Polygon ID Edge ID Vertex FROM ID Vertex TO ID 0 0 0 1 0 1 1 5 0 2 5 6
[0072] The polygon ID, vertex ID, and vertex type are some of the attributes that can be used to describe and manipulate polygons. Polygon ID may be a unique identifier that may be assigned to each polygon **318**C in an image. The polygon ID may be used to access, modify, or delete a specific polygon from a collection of polygons. For example, if an image contains multiple polygons, each polygon can be given a different ID, such as 1, 2, or 3. One can use the ID to select a polygon and change its color, position, or shape. The vertex ID may be a unique identifier that may be assigned to each vertex in a polygon. The vertex ID may be used to access, modify, or delete a specific vertex from a polygon. For example, if a polygon has four vertices, each vertex can be given a different ID, such as 1, 2, 3, or 4. One can use the ID to select a vertex and change its coordinates or add or remove edges from it. Vertex type may be a classification that indicates the role or function of a vertex in a polygon. Vertex type can be used to determine how a polygon is drawn, filled, or textured. For example, some common vertex types include a corner vertex that forms a sharp angle between two edges, a curve vertex that forms a smooth curve between two edges, a texture vertex that specifies the texture coordinates for a polygon, and the like. A 3D image **318**A (obtained after polygon shaping) includes 3D polygons **318**C on a ground segment **318**B, which are shaped based on the correspondence table **316**A.
[0073] At **320**, a projection of 3D polygons **318**C may be performed. The 3D polygons **318**C may be projected onto a ground segment **318**B of the projected 3D surface model **320**A based on the correspondence information corresponding to one or more objects. Further, the projection may be performed further based on the vertex information and edge information. The polygon projection is described in FIG. **4**, FIG. **5**, FIG. **7**, and FIG. **9**.
[0074] FIG. **4** is a diagram that illustrates a flowchart of an example method for estimation of 3D polygons corresponding to objects in 2D images **110**A, in accordance with an embodiment of the disclosure. FIG. **4** is described in conjunction with elements from FIG. **1**, FIG. **2**, and FIG. **3**. With reference to FIG. **4**, there is shown a flowchart **400**. The method illustrated in the flowchart **400** may start at **402** and may be performed by any suitable system, apparatus, or device, such as, by the example electronic device **102** of FIG. **1**, or the circuitry **206** of FIG. **2**. Although illustrated with discrete blocks, the steps and operations associated with one or more of the blocks of the flowchart **400** may be divided into additional blocks, combined into fewer blocks, or eliminated, depending on the implementation.
[0075] At block **402**, historical data associated with the 3D polygons corresponding to the objects in the 2D images **110**A may be retrieved. The historical data may be stored in the database **106**. Each 3D polygon may enclose a corresponding object of the plurality of objects. The historical data collection may be done through various known methods to examine historical images stored in the database **106**, such as objects captured in the 2D images **110**A or 3D images across various timeframes, different angles, and the like.
[0076] At block **404**, polygon estimation model **208**B may be trained based on the historical data.
[0077] At block **406**, the 3D polygons may be estimated corresponding to detected objects. The objects may be detected based on the 3D polygons on the ground region of the 2D images **110**A. The prediction of the behavioral data may involve use of machine learning to analyze how behavioral data of polygons in different scenarios, such as moving, rotating, or scaling. The polygon estimation model **208**B may include a reinforcement learning (RL) framework, where an agent learns to control the polygons by interacting with an environment and receiving rewards or penalties based on the actions of the objects (corresponding to the 3D polygons).
[0078] The polygon estimation model **208**B may optimize the behavioral data of the objects for

different objectives, such as but not limited to, reaching a target, avoiding obstacles, or following a path. In an embodiment, a generative adversarial network (GAN) may be used to determine the behavioral data of the objects corresponding to the 3D polygons, where a generator tries to estimate a polygon behavior and a discriminator tries to distinguish between real and generated behaviors. Based on the behavioral data the estimation of 3D polygons may be performed. The generator may learn to generate diverse and natural polygon behaviors that match the data distribution. The predefined threshold value may vary based on the past instances of the objects behavior and may represent physical laws and human movements. For example, a predefined threshold value may be used to determine whether the object detector **208**A has correctly identified an object in an image, based on the intersection over union (IoU) metric.

[0079] At block **408**, a difference between the estimated 3D polygons corresponding to the detected objects may be estimated. When the difference between the estimated 3D polygons and the detected 3D polygons is less than a predefined threshold, then control may pass to **410**. When the difference between the estimated 3D polygons and the detected 3D polygons is more than a predefined threshold, then control may pass to **412**. The polygons may be further estimated to optimize the object detection, as the detected polygons may be inaccurate due to occlusions. Therefore, the estimated 3D polygons corresponding to the objects and the detected 3D polygons may be compared to determine a behavioral difference.

[0080] The threshold value may depend on the movement of objects, rotation, scales, or change of shape in previous images or videos. For example, when the object is a car, the threshold value may be higher when the car is moving fast or turning sharply, and lower when the car is moving slowly or staying still. This means that the threshold value may be based on rules or patterns that describe how the object behaves in the real world. For example, if the object is a person, the threshold value may be based on the human anatomy, posture, gesture, or facial expression. If the object is for example a ball, the threshold value may be based on the physics of motion, gravity, or friction. The projection of polygons is described in detail in exemplary FIG. **5**, and FIG. **9**.

[0081] At block **410**, the detected 3D polygons may be projected onto the ground region of the 2D images **110**A based on a determination that a difference between the estimated 3D polygons and the detected 3D polygons is less than a predefined threshold. Additionally, or alternatively, the projection of the detected 2D polygons may be used to project the polygons (or 3D polygons) onto the 3D surface model **112**A.

[0082] In an embodiment, the detected 3D polygons may be dynamically detected based on the 2D images **110**A. The 3D polygons estimated based on the historical data may be referred as the estimated polygons. The 3D polygons may include a depth in addition to other 2D parameters. The polygons may be estimated even if detected polygons can be obtained from the dynamic 2D image **110**A because detected polygon may be inaccurate or missing due to the occlusions. In case of a human object, the estimated polygons may be learned from past instances and may broadly represent physical laws and human movement.

[0083] At block **412**, the estimated 3D polygons may be projected onto the ground region of the 2D images **110**A based on a determination that a difference between the estimated 3D polygons and the detected 3D polygons is more than a predefined threshold. Additionally, or alternatively, the projection of the estimated 3D polygons may be used to project the 3D polygons (or polygons) onto the 3D surface model **112**A.

[0084] The estimated polygons may be estimated even if detected 3D polygons may be obtained from the dynamic 2D images **110**A because detected 2D polygons may be inaccurate due to the occlusion. The estimated 3D polygons may be learned from past instances and broadly represent physical laws and human movement. Therefore, it is assumed that they will not deviate significantly compared to detected polygons based on fragmented information.

[0085] FIG. **5** is a diagram that illustrates an exemplary scenario for estimation of 3D polygons, in accordance with an embodiment of the disclosure. FIG. **5** is described in conjunction with elements from FIG. **1**, FIG. **2**, FIG. **3**, and FIG. **4**. With reference to FIG. **5**, there is shown the exemplary scenario **500**. The method illustrated in the exemplary scenario **500** may be performed by any suitable system, apparatus, or device, such as, by the example electronic device **102** of FIG. **1** or circuitry **206** of FIG. **2**.

[0086] Consider a set of image frames **504**, **506**, **508**, and **510** (e.g., 2D images) as examples for determining behavioral data of the objects **504**D across the set of frames **504**, **506**, **508**, and **510**. The number of image frames shown in FIG. **5** is presented merely as an example and such an example should not be construed as limiting the disclosure. The image frames **504**, **506**, **508**, and **510** may include only one image frame or more than N image frames for determining the behavioral data, without deviation from the scope of the disclosure. For the sake of brevity, only four image frames are shown in FIG. **5**. However, in some embodiments, there may be more than four image frames, without limiting the scope of the disclosure.

[0087] At frame **504**, an object **504**D may be detected. The number of objects shown in FIG. **5** is presented merely as an example. The object **504**D may include only one object or more than one object, without deviation from the scope of the disclosure.

[0088] The first step is to determine the number of objects **504**D within the set of image frames **504**, **506**, **508**, and **510**. The set of image frames **504**, **506**, **508**, and **510** may be analyzed to determine the object of interest (e.g., the object **504**D) or objects to be tracked for the behavioral data. The dynamic objects may be considered as the object of interest in the image frames **504**, **506**, **508**, and **510** for determining the behavioral data in some embodiments. For the object **504**D detected within the image frame **504**, a 3D polygon **504**C may be estimated at time ‘t’. The image frame **504** at time ‘t’ may include multiple objects. For example, the image frame **504** may include a ground segment and the object **504**D to be tracked for the behavioral data. The object **504**D may be at a location **1** (the object may be enclosed by the 3D polygon **504**C) in the image frame **504** at the time t. Consider the image frame **506** at time t+1, where image frame **506** may include occlusions **504**A and **504**B. The object **504**D may now be detected at a second location in the second image frame **506**, by considering behavioral data of the object. The polygon estimation model **208**B may be pretrained using the historical data of the objects and may compensate the lack of polygons due to non-detection and inaccurate objects (at locations of the occlusions **504**B, **504**A in image frames **506**, **508**). The polygon ID, vertex ID, and vertex type are some of the attributes that can be used to describe and manipulate polygons. In an embodiment, each polygon may include vertices and edges and the information related to the vertices and edges may be stored in the form of table (Table. **1** and Table. **2** of FIG. **3**). The table may include vertex information and the edge information.

[0089] In the image frame **508**, the object **504**D may be hidden behind the occlusion **504**A and **504**B at time t+j. The image frame **510** at time t+j+1 may be considered where the object **504**D may be detected at time t+j+1. The object **504**D corresponding to the 3D polygon **504**C may be estimated for image frames **506** and **508** where the object **504**D is not visible due to the occlusions **504**B, **504**A. The location of the estimated object **504**D may be determined based on the behavioral data.

[0090] FIG. **6** is a diagram that illustrates an exemplary scenario for acquisition of 3D surface model, in accordance with an embodiment of the disclosure. FIG. **6** is described in conjunction with elements from FIG. **1**, FIG. **2**, FIG. **3**, FIG. **4**, and FIG. **5**. With reference to FIG. **6**, there is shown an exemplary scenario **600**. The method illustrated in the exemplary scenario **600** may be performed by any suitable system, apparatus, or device, such as, by the example electronic device **102** of FIG. **1** or the circuitry **206** of FIG. **2**.

[0091] The scenario **600** may include the 3D imaging device **112**. The 3D imaging device **112** may include suitable logic, circuitry, or interfaces, which may be configured to capture the 3D view of the real-world location (e.g., a street in a city). In accordance with an embodiment, the 3D imaging device **112** may include a plurality of image sensors (not shown) to capture the 3D view of the real-world location from multiple viewpoints. In accordance with an embodiment, the 3D imaging device **112** may be integrated into the electronic device **102**. Examples of the 3D imaging device **112** may include, but are not limited to, LiDAR, a time-of-flight camera (ToF camera), stereoscopic camera system, structured-light 3D scanner, or CT scanner.

[0092] In an embodiment, the 3D imaging device **112** may be a LIDAR camera. The LiDAR camera may work by sending out rapid pulses of laser light and detecting the reflected light with a sensor. By scanning the laser beam across the scene, the LiDAR camera may create a 3D point cloud **602**A of an object or a scene **602**, which can be further processed into a 3D surface model image **604**. For example, the 3D point cloud **602**A may be converted to a surface model or mesh by application of a meshing operation on the 3D point cloud **602**A. The 3D surface model image **604** may be referred to as a geometric data most often composed of a bunch of connected triangles (or polygons) that explicitly describe a surface. The 3D surface model image **604** may consist of vertices, edges, and faces that define the surface of the object. To generate the 3D surface model image **604** from the 3D point cloud **602**A, different methods and algorithms, such as Poisson surface reconstruction, ball-pivoting algorithm, Delaunay triangulation, or marching cubes may be used.

[0093] FIG. **7** is a diagram that illustrates a flowchart of an example for determination of correspondence information for dynamic multi-dimensional media content projection, in accordance with an embodiment of the disclosure. FIG. **7** is described in conjunction with elements from FIG. **1**, FIG. **2**, FIG. **3**, FIG. **4**, and FIG. **5**. With reference to FIG. **7**, there is shown an exemplary flowchart **700**. The method illustrated in the exemplary flowchart **700** may be performed by any suitable system, apparatus, or device, such as, by the example electronic device **102** of FIG. **1**, or the circuitry **206** of FIG. **2**. Although illustrated with discrete blocks, the steps and operations associated with one or more of the blocks of the flowchart **700** may be divided into additional blocks, combined into fewer blocks, or eliminated, depending on the particular implementation.

[0094] At **704**, an image size of the 2D images **110**A may be extracted. The 2D images **110**A may be provided as input for performing the image extraction. The 2D image **110**A may be retrieved from the database **106** and the image size may be extracted based on dimensions of an image in terms of pixels, such as height and width. In another embodiment, image size may be extracted by dividing 2D images **110**A into smaller patches of a fixed size and using the dimensions of each patch to extract the image size.

[0095] At **706**, a 2D coordinate of a pixel from 2D pixel coordinates of the 2D images **110**A may be selected. The 2D pixel coordinates may be considered as X.sub.2D, Y.sub.2D, wherein X.sub.2D is a position of a pixel along an x-axis and Y.sub.2D is a position of the pixel along a y-axis. As an example, the 2D coordinates of two pixels (ID1_2D and ID2_2D) may be represented as (x,y)=(**305**,**656**), (x,y)=(**506**, **656**)

[0096] At **708**, ray casting operation may be applied on the selected 2D coordinates to estimate a corresponding 3D surface coordinate on a ground segment of the 3D surface model **112**A in the 3D space. The 3D surface coordinates may include the 3D pixel coordinates (X.sub.3D, Y.sub.3D, Z.sub.3D). As an example, the 3D coordinates for the two pixels ID1_2D and ID2_3D are given below:

[00001] $(x, y, z) = (0.1, 0.02, 0.21); (x, y, z) = (0.21, 0.5, 0.1);$ .Math.

[0097] At **710**, correspondence information (e.g., the correspondence table **316**A) may be determined based on 2D pixel coordinates and 3D surface coordinate. The 2D coordinate selection from 2D pixel coordinates may be a process of converting the pixel locations of the 2D image **110**A to a different coordinate system, such as world coordinates, camera coordinates, or intrinsic coordinates. This can be useful for various applications, such as 3D surface model construction, image registration, or feature extraction.

[0098] At **712**, it may be determined whether the selected 2D coordinate is the final coordinate of the 2D image. If the selected 2D coordinate is the final coordinate of the 2D image, control may pass to end. Otherwise, the control may pass to **706** and the next 2D coordinate from the 2D image may be selected. The process from **706** to **710** may be repeated for all the pixels of the 2D image.

[0099] FIG. **8** is a diagram that illustrates an exemplary scenario for estimation of initial camera pose parameters corresponding to an initial view of the 3D surface model in a 3D space, in accordance with an embodiment of the disclosure. FIG. **8** is described in conjunction with elements from FIG. **1**, FIG. **2**, FIG. **3**, FIG. **4**, FIG. **5**, FIG. **6**, and FIG. **7**. With reference to FIG. **8**, there is shown an exemplary scenario **800**. In the exemplary scenario **800**, there is shown an initial view **802** of the 3D surface model, a final view **804** of the 3D surface model (after alignment), and a base 2D image **806**. To align the initial view **802** of the 3D surface model with a view of the base 2D image **806**, the circuitry **206** may estimate initial camera pose parameters corresponding to the initial view **802** of the 3D surface model in a 3D space. Thereafter, the circuitry **206** may iteratively optimize the initial camera pose parameters to align the initial view **802** of the 3D surface model to the view of the base 2D image **806**. After the alignment, the final view **804** of the 3D surface model may match the view of the base 2D image **806**.

[0100] FIG. **9** is a diagram that illustrates a flowchart of an example for dynamic multi-dimensional media content projection, in accordance with an embodiment of the disclosure. FIG. **9** is explained in conjunction with elements from FIG. **1**, FIG. **2**, FIG. **3**A, FIG. **3**B, FIG. **4**, FIG. **5**, FIG. **6**, FIG. **7**, and FIG. **8**. With reference to FIG. **9**, there is shown a flowchart **900**. The operations from **902** to **916** may be implemented by any computing system, such as, by the electronic device **102**, or the circuitry **206** of the electronic device **102** of FIG. **1**. The operations may start at **902** and may proceed to **904**.

[0101] At **904**, image data including the 2D images **110**A depicting view of real-world location may be acquired. The 2D images **110**A may be any digital data, which can be rendered, streamed, broadcasted, or stored on any electronic device or storage. Examples of the 2D images **110**A may include, but are not limited to, images (such as overlay graphics), real world images captured by imaging devices, or audio/video data.

[0102] At **906**, the 3D polygons corresponding to objects may be detected on a ground region of the 2D images **110**A. The polygons may be 3D polygons with straight sides that form a closed area. The polygons may include a cube, a cuboid, and so on. The vertices, edges, and faces of these polygons may be defined using points (Pt) with x, y, and z coordinates.

[0103] The polygon generation may be a process that involves creating a set of points to define the boundaries of the objects or region of interest in the images or videos. The points, or vertices may be connected by straight lines to form a polygon that surrounds the object or region, taking its shape. For example, the vertices may be generated from bounding box coordinates by selecting four points on a perimeter of a rectangular bounding box and using such points as the polygon vertices.

[0104] At **908**, 3D surface model **208**C of real-world location may be acquired. The 3D surface model **112**A may be a digital representation of features in three-dimensional space. Some examples of 3D surfaces may be landscape, an urban corridor, gas deposits under the earth, and a network of well depths and the like to determine water table depth. The 3D surface model **112**A may be created from a variety of data sources, such as points, lines, polygons, or images, using interpolation or triangulation methods.

[0105] In an embodiment, initial camera pose parameters may be estimated corresponding to an initial view of the 3D surface model **208**C in a 3D space. The initial camera pose parameters may be optimized to align a view of the 3D surface model **208**C to a view of the 2D images **110**A.

[0106] At **910**, the 3D surface model **112**A may be segmented into 3D segments including ground segment and one or more non ground segments. In some embodiments, various techniques, and algorithms, such as but not limited to, thresholding, clustering, edge detection, region growing, graph-based methods, or deep learning models may be used to perform model segmentation.

Additionally, or alternatively, the segmentation may be performed by acquiring the 2D images **110**A having multiple ground segments such as roads and non-ground segments such as buildings. The 2D images **110**A may be collected for example, traffic surveillance images featuring the multiple objects. The collected 2D images **110**A may be annotated. This involves drawing bounding boxes around the objects of interest in each 2D image **110**A and labeling them. The objects of interest may be, for example, ground segments **312**B. For instance, a bounding box may be drawn around the ground segment and labeled as 'ground'. The annotated 2D images **110**A may be used to train a machine learning model. The model may learn to recognize the features of each object from the annotated images.

[0107] In an embodiment, the acquisition of the 3D surface model **112**A may include capturing of 3D data via image capturing device to generate a 3D point cloud based on the 3D data. The point cloud data may be converted into the 3D surface model **208**C.

[0108] At **912**, ground segment of 3D surface model **112**A may be extracted by removing one or more non-ground segments from the plurality of 3D segments.

[0109] At **914**, correspondence information between 2D pixel coordinates in 2D images **110**A and 3D surface coordinates of the ground segment may be determined by applying ray casting operation on 2D images **110**A and the ground segment of 3D surface model **112**A. The ray casting operation may be applied on the selected 2D coordinates to estimate corresponding 3D surface coordinates on the ground segment in the 3D space.

[0110] In an example embodiment, the correspondence information may be a table **316**A that includes the 2D pixel coordinates and the 3D surface coordinates corresponding to the 2D pixel coordinates. The correspondence table **316**A may be useful to quickly transfer the polygon data from 2D to 3D coordinates on the ground segment of the 3D surface model **208**C.

[0111] At **916**, one or more 3D polygons corresponding to objects may be projected onto the 3D surface model **208**C based on correspondence information. In an embodiment, each 3D polygon of the one or more 3D polygons may enclose a corresponding object of the plurality of objects in the 3D space of the 3D surface model **208**C. Vertex information including a mapping of the 3D surface coordinates with vertices of the one or more 3D polygons may be generated. Edge information may be generated to include a mapping of the vertices to edges of the one or more 3D polygons. The projection may be performed further based on the vertex information and the edge information. The vertex information may include a primary table that includes a vertex ID, a polygon ID, a vertex type, and the 3D surface coordinates, and the edge information may include a secondary table that includes the polygon ID, an edge ID, and IDs of edge vertices. The vertex type may include one of a ground or an object.

[0112] In an embodiment, the projection of the polygons may include retrieving historical data associated with the 3D polygons corresponding to the objects in of the 2D images **110**A or 3D surface model **112**A. Each 3D polygon encloses a corresponding object of the plurality of objects. The polygon estimation model **208**B may be trained based on the historical data. Based on the trained polygon estimation model **208**B, 3D polygons may be estimated corresponding to the detected objects in 2D images **110**A. The detected 3D polygons may be projected onto the ground region of the 2D images **110**A based on the determination that the difference between the estimated 3D polygons and the detected 3D polygons is less than the predefined threshold. The estimated 3D polygons may be projected onto the ground region of the 2D images **110**A based on the determination that a difference between the estimated 3D polygons and the detected 3D polygons is greater than the predefined threshold.

[0113] A person with ordinary skill in the art will understand that the scope of the disclosure is not limited to 2D images/3D images. In accordance with an embodiment, 2D and 3D images may be any data such as videos, moving pictures and the like. without departure from the scope of the disclosure.

[0114] Various embodiments of the disclosure may provide one or more non-transitory computer-readable storage media configured to store instructions that, in response to being executed, cause a system (such as, the example electronic device **102**) to perform operations. The operations may include acquiring image data (for example, 2D image data) that includes two-dimensional (2D) images depicting a view of a real-world location (for example, a live recording received from a CCTV camera). The operations may comprise detecting one or more 2D polygons corresponding to one or more objects on a ground region of the 2D images and acquiring a 3D surface model of the real-world location. The set of operations may further include segmenting the 3D surface model into a plurality of 3D segments comprising a ground segment and one or more non-ground segments from the plurality of 3D segments. The set of operations may further comprise determining, by applying a ray casting operation on the 2D images and the ground segment of the 3D surface model, correspondence information between 2D pixel coordinates in the 2D images and 3D surface coordinates of the ground segment, and projecting, based on the correspondence information, one or more 3D polygons corresponding to the one or more objects onto the 3D surface model.

[0115] As used in the present disclosure, the terms “module” or “component” may refer to specific hardware implementations configured to perform the actions of the module or component and/or software objects or software routines that may be stored on and/or executed by general purpose hardware (e.g., computer-readable media, processing devices, etc.) of the electronic device **102**. In some embodiments, the different components, modules, engines, and services described in the present disclosure may be implemented as objects or processes that execute on the electronic device **102** (e.g., as separate threads). While some of the system and methods described in the present disclosure are described as being implemented in software (stored on and/or executed by general purpose hardware), specific hardware implementations or a combination of software and specific hardware implementations are also possible and contemplated. In this description, a “computing entity” may be any electronic device **102** as previously defined in the present disclosure, or any module or combination of modulates running on the electronic device **102**.

[0116] Additionally, if a specific number of an introduced claim recitation is intended, such an intent will be explicitly recited in the claim, and in the absence of such recitation no such intent is present. For example, as an aid to understanding, the following appended claims may contain usage of the introductory phrases “at least one” and “one or more” to introduce claim recitations. However, the use of such phrases should not be construed to imply that the introduction of a claim recitation by the indefinite articles “a” or “an” limits any particular claim containing such introduced claim recitation to embodiments containing only one such recitation, even when the same claim includes the introductory phrases “one or more” or “at least one” and indefinite articles such as “a” or “an” (e.g., “a” and/or “an” should be interpreted to mean “at least one” or “one or more”); the same holds true for the use of definite articles used to introduce claim recitations.

[0117] Further, any disjunctive word or phrase presenting two or more alternative terms, whether in the description, claims, or drawings, should be understood to contemplate the possibilities of including one of the terms, either of the terms, or both terms. For example, the phrase “A or B” should be understood to include the possibilities of “A” or “B” or “A and B.”

[0118] All examples and conditional language recited in the present disclosure are intended for pedagogical objects to aid the reader in understanding the present disclosure and the concepts contributed by the inventor to furthering the art and are to be construed as being without limitation to such specifically recited examples and conditions. Although embodiments of the present disclosure have been described in detail, various changes, substitutions, and alterations could be made hereto without departing from the spirit and scope of the present disclosure.

Claims

1. A method, executed by a processor in an electronic device, comprising: acquiring image data that includes two-dimensional (2D) images depicting a view of a real-world location; detecting one or more 3D polygons corresponding to one or more objects on a ground region of the 2D images; acquiring a three-dimensional (3D) surface model of the real-world location; segmenting the 3D surface model into a plurality of 3D segments comprising a ground segment and one or more non-ground segments; and extracting the ground segment of the 3D surface model by removing the one or more non-ground segments from the plurality of 3D segments; determining, by applying a ray casting operation on the 2D images and the ground segment of the 3D surface model, correspondence information between 2D pixel coordinates in the 2D images and 3D surface coordinates of the ground segment; and projecting, based on the correspondence information, the one or more 3D polygons corresponding to the one or more objects onto the 3D surface model.
2. The method according to claim 1, further comprising: estimating initial camera pose parameters corresponding to an initial view of the 3D surface model in a 3D space; and optimizing the initial camera pose parameters to align a view of the 3D surface model to a view of the 2D images.
3. The method according to claim 1, wherein the correspondence information includes a table that includes the 2D pixel coordinates and the 3D surface coordinates corresponding to the 2D pixel coordinates.
4. The method according to claim 1, further comprising: generating vertex information that includes a mapping of the 3D surface coordinates with vertices of the one or more 3D polygons; and generating edge information that includes a mapping of the vertices to edges of the one or more 3D polygons; wherein the projection of the one or more 3D polygons is performed further based on the vertex information and the edge information.
5. The method according to claim 4, wherein the vertex information includes a primary table that includes a vertex ID, and a polygon ID, a vertex type, and the 3D surface coordinates, and the edge information includes a secondary table that includes the polygon ID, an edge ID, and IDs of edge vertices.
6. The method according to claim 4, wherein the vertex type includes one of a ground or an object.
7. The method according to claim 1, wherein each 3D polygon of the one or more 3D polygons encloses a corresponding object of the plurality of objects in a 3D space of the 3D surface model.
8. The method according to claim 1, further comprising: retrieving, from a database, historical data associated with a plurality of 2D polygons corresponding to a plurality of objects in a set of images, wherein each 3D polygon of the plurality of 3D polygon encloses a corresponding object of the plurality of objects; training a polygon estimation model based on the historical data; estimating one or more 3D polygons corresponding to the detected one or more objects in the 2D images, based on the trained polygon estimation model; and projecting the detected one or more 3D polygons onto the ground region of the 2D images based on a determination that a difference between the estimated one or more 3D polygons and the detected one or more 3D polygons is less than a predefined threshold.
9. The method according to claim 8, further comprising projecting the estimated one or more 3D polygons onto the ground region of the 2D images based on a determination that a difference between the estimated one or more 3D polygons and the detected one or more 3D polygons is greater than the predefined threshold.
10. The method according to claim 1, wherein the determining of the correspondence information includes: extracting an image size of the 2D images; selecting a 2D coordinate from 2D pixel coordinates; applying the ray casting operation on the selected 2D coordinate to estimate a corresponding 3D surface coordinate of the 3D surface coordinates on the ground segment in a 3D space; and determine the correspondence information based on the 2D pixel coordinate and the 3D surface coordinate.
11. The method according to claim 1, wherein the acquiring of the 3D surface model further

includes: capturing 3D data via an image capturing device; generating a 3D point cloud based on the 3D data; and converting the point cloud data into the 3D surface model.

12. One or more non-transitory computer-readable storage media configured to store instructions that, in response to being executed, cause an electronic device to perform operations, the operations comprising: acquiring image data that includes two-dimensional (2D) images depicting a view of a real-world location; detecting one or more 3D polygons corresponding to one or more objects on a ground region of the 2D images; acquiring a three-dimensional (3D) surface model of the real-world location; segmenting the 3D surface model into a plurality of 3D segments comprising a ground segment and one or more non-ground segments; and extracting the ground segment of the 3D surface model by removing the one or more non-ground segments from the plurality of 3D segments; determining, by applying a ray casting operation on the 2D images and the ground segment of the 3D surface model, correspondence information between 2D pixel coordinates in the 2D images and 3D surface coordinates of the ground segment; and projecting, based on the correspondence information, the one or more 3D polygons corresponding to the one or more objects onto the 3D surface model.

13. The one or more non-transitory computer-readable storage media according to claim 12, wherein the operations further comprises: estimating initial camera pose parameters corresponding to an initial view of the 3D surface model in a 3D space; and optimizing the initial camera pose parameters to align a view of the 3D surface model to a view of the 2D images.

14. The one or more non-transitory computer-readable storage media according to claim 12, wherein the operations further comprises: generating vertex information that includes a mapping of the 3D surface coordinates with vertices of the one or more 3D polygons; generating edge information that includes a mapping of the vertices to edges of the one or more 3D polygons, wherein the projection is performed further based on the vertex information and the edge information.

15. The one or more non-transitory computer-readable storage media according to claim 14, wherein the vertex information includes a primary table that includes a vertex ID, and a polygon ID, a vertex type, and the 3D surface coordinates, and the edge information includes a secondary table that includes the polygon ID, an edge ID, and IDs of edge vertices.

16. The one or more non-transitory computer-readable storage media according to claim 12, further comprising: retrieving, from a database, historical data associated with a plurality of 3D polygons corresponding to a plurality of objects in a set of images, wherein each 3D polygon of the plurality of 3D polygon encloses a corresponding object of the plurality of objects; training a polygon estimation model based on the historical data; estimating one or more 3D polygons corresponding to the detected one or more objects in the 2D images, based on the trained polygon estimation model; and projecting the detected one or more 3D polygons onto the ground region of the 2D images based on a determination that a difference between the estimated one or more 3D polygons and the detected one or more 3D polygons is less than a predefined threshold.

17. The one or more non-transitory computer-readable storage media according to claim 12, wherein the operations further comprises: extracting an image size of the 2D images; selecting a 2D coordinate from 2D pixel coordinates; applying the ray casting operation on the selected 2D coordinate to estimate a corresponding 3D surface coordinate of the 3D surface coordinates on the ground segment in a 3D space; and determining the correspondence information based on the 2D pixel coordinate and the 3D surface coordinate.

18. The one or more non-transitory computer-readable storage media according to claim 12, wherein the operations further comprises: capturing 3D data via an image capturing device; generating a 3D point cloud based on the 3D data; and converting the point cloud data into the 3D surface model.

19. An electronic device, comprising: a memory configured to store instructions; and a processor, coupled to the memory, configured to execute the instructions to perform a process comprising:

acquiring image data that includes two-dimensional (2D) images depicting a view of a real-world location; detecting one or more 3D polygons corresponding to one or more objects on a ground region of the 2D images; acquiring a three-dimensional (3D) surface model of the real-world location; segmenting the 3D surface model into a plurality of 3D segments comprising a ground segment and one or more non-ground segments; extracting the ground segment of the 3D surface model by removing the one or more non-ground segments from the plurality of 3D segments; determining, by applying a ray casting operation on the 2D images and the ground segment of the 3D surface model, correspondence information between 2D pixel coordinates in the 2D images and 3D surface coordinates of the ground segment; and projecting, based on the correspondence information, the one or more 3D polygons corresponding to the one or more objects onto the 3D surface model.